

Week 7

Azure Cloud Fundamentals for ML and VM Provisioning

- Understand cloud computing models in ML environments.
- Explore Azure services suitable for machine learning.
- Learn how to select and provision GPU-enabled virtual machines.
- Prepare for practical deployment of ML workflows in cloud.

Learning Objectives

- Describe key Azure services for ML and AI.
- Configure Azure accounts; set up resource groups and permissions.
- Compare VM options for ML workloads; focus on GPU VMs.
- Gain practical skills in provisioning and accessing cloud VMs.
- Align operations with industry ML engineering standards.

Agenda

- Cloud models for ML workloads.
- The Azure landscape for AI and data science.
- Configuring resource groups and permissions in Azure.
- Comparing and selecting Azure VM types.
- Hands-on activity; Provisioning a GPU VM.
- Industry practices; Security and model management.
- Preparing for practical assessment.

Cloud Computing Models for ML

- Three main models; Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).
- IaaS provides raw compute and storage; maximum control for ML projects.
- PaaS offers managed frameworks; focus on code not infrastructure.
- SaaS delivers ready-made ML tools; limited customization.
- ML and AI engineering often combine IaaS and PaaS for flexibility and scalability.

Azure Environment for ML Workloads

- Azure offers a broad ecosystem; compute, storage, and ML tools.
- Key services; Azure Machine Learning, Azure Databricks, and Azure Virtual Machines.
- Data scientists and engineers often use Azure Notebooks and Azure Storage for collaborative workflows.
- Azure integrates with open-source ML frameworks such as PyTorch.

Resource Groups and Account Structure

- Resource groups organize Azure assets for easier management.
- Each ML project should use a separate group; supports cost tracking and security.
- Permissions in Azure are structured; subscriptions contain resource groups which contain resources.
- Use tags to label resources for compliance and reporting.

User Access and Permissions in Azure

- Role-based access control (RBAC) assigns users predefined roles.
- Principle of least privilege; give users only the permissions necessary for their tasks.
- Common roles; Owner, Contributor, Reader, and specific roles for VM, network, and storage.
- Always secure access with multifactor authentication.

Virtual Machines for ML Workloads

- VMs provide scalable compute power for ML training and inference.
- Key options; CPU-only VMs, standard GPU VMs, high performance (HPC) VMs.
- GPU VMs significantly accelerate PyTorch training and model experimentation.
- VM choice impacts speed, cost, and reliability of ML workflows.

Comparing Azure Compute Options

VM Type	Use Case	GPU Enabled	Example
Standard CPU VM	Scripting, basic workloads	No	B2s, D2s_v3
Standard GPU VM	ML training, data science	Yes	NC, ND, NV series
HPC VM	Large-scale, distributed ML	Yes	HB, HC series

- Select resources based on job size, cost, and model complexity.

Practical Steps; Provisioning a GPU VM

1. Log in to Azure portal.
2. Navigate to Resource Groups and create a new group for your ML project.
3. Choose 'Create Resource', then select Virtual Machine.
4. Select a GPU-enabled VM (e.g. NC series).
5. Set basic settings; region, admin username, public access.
6. Configure advanced options; networking, security groups.
7. Review and launch the VM.
8. Connect via SSH with generated keys.

Example Scenario; Training ML Models on Azure GPU VM

- Data scientist provisions an NC6 VM in Azure for training a vision model in PyTorch.
- Data is uploaded to Azure storage; VM connects for data access.
- Training scripts use CUDA to accelerate learning.
- Upon completion, models and logs are saved to storage for retrieval and monitoring.

Industry Standards; Practices and Pitfalls

- Use version control for scripts and infrastructure settings.
- Monitor VM use; shut down when idle to avoid unexpected costs.
- Apply tags for tracking project spend and asset ownership.
- Always comply with responsible AI and security standards on the cloud.

In-Class Activity; Exploring Azure Portal

- Log in to Azure portal with provided credentials.
- Create a new resource group for a sample ML project.
- Browse available VM sizes; compare cores, memory, GPU capability.
- Practice setting permissions for a classmate as Contributor.

Connecting to Course Progression

- Builds directly on Bash scripting and PyTorch skills from previous weeks.
- Prepares foundation for deploying ML code on cloud-based infrastructure.
- Enables transition to distributed training and high-performance workflows in later weeks.

Assessment Preparation

- Document the steps taken to create and access a GPU VM in Azure.
- Take and save screenshots at key stages for portfolio submission.
- Reflect on resource selection; justify your VM choice in simple terms.
- Prepare for technical demonstration in future weeks.

Summary and Next Steps

- Azure cloud is foundational for scalable, production-ready ML.
- Understanding provisioning, access, and resource management is key.
- Next week; Deep dive into cloud networking and storage for AI workflows.
- Remember to review Azure portal navigation and account management skills.

